

Course Project Requirements

Theoretical/Practical study

Computer Vision (Visión Artificial) Course 2025-2026

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1. Subject, project parts, and scope

The subject project will be completed in groups of -minimum- three. Each group will decide to do a project on these three different approaches:

- **Reto Michelin** (see individual PDF). Solve a real problem for the company Michelin.
- Work on a specific problem of subject block or computer vision task: segmentation, classification, object detection, and recognition.
- Choose **one topic** from the proposed list, or the groups could suggest another one that needs to be approved by the instructor. List of projects proposed as examples in this academic course:
 - Smishing: Text spotting (detection and recognition)
 - Face detection
 - Age estimation
 - Animal monitoring in industry or animal welfare on farms.

At least 50% of the students should voluntarily participate in the Reto Michelin. These students will have slightly different objectives, but the same deliverables. If there are not 50% of volunteers, the professor will randomly select groups to participate in.

The objectives of the project are:

- Research the problem selected and present your findings to your colleagues.
- Solve the problem using traditional and deep learning techniques and decide which one is best.

Two groups can work on the same problem/project, but in such cases, each group's datasets must be different. To make the selection, groups will have one week to register and another week to select the project they will **work on**, both in a dedicated forum (**Course Project Groups**).

The project will be divided into the following **BLOCKS**. Between brackets, the punctuation assigned to each part:

- Research proposal & presentation **[12 points]**
- Proof of concept & written evaluation **[18 points]**

The following sections will go into detail about each block of the project.

2. Block 01: Research Proposal and Presentation

The group can prepare the research proposal in LaTeX using the template "GIDIA_VA_Template_Research_Proposal.zip" in Agora (*recommended for use on the Overleaf platform*). Alternatively, the group can work in Word or another file editor, but they should create their template based on the provided PDF.

The group must deliver the pdf containing this research proposal in Agora before the specified date **(18/03/2026)**. Based on the content provided, the group might receive small written feedback before the presentation.

The theoretical study will include the following sections.

2.1. Introduction

The introduction will briefly describe the problem, why it is important, what others have been doing, the solution proposed, and the challenges...

In other words, be more specific about what you need to do 😊 Try to reply to the questions inserted in the template with the text you would insert, and you could get a piece of text to serve as an introduction.

2.2. Proposed Solutions

The group must review no fewer than **10 papers** from the literature, **per group member**. Papers reviewed must be "recent" and not more than three years old (*i.e., select papers from 2024, 2025, 2026*). Below is an example of the level of detail expected for each method reviewed.

Example of paper/method reviewed for a research line related to Phishing detection: \textit{Adebawale et al. \cite{Adebawale2019} created a browser extension to protect users by extracting features from the URL, the source code, the images, and third-party services like WHOIS. Those features were introduced into an ANFIS (Adaptive Neuro-Fuzzy Inference System) and combined with SIFT (Scale-Invariant Feature Transform), obtaining an accuracy of 98.30\% on Rami et al. dataset\footnote{\url{http://eprints.hud.ac.uk/24330/9/Mohammad14JulyDS}}, collected in 2015.}

A table summarizing the main methods would be required (voluntary in the research proposal, mandatory in the presentation). An example of this kind of table appears below in Figure 1, where methods for face detection are analyzed. As can be seen, the average precision, or another performance metric used to evaluate the technique, must be provided.

Method	Year	Description	Evaluation Dataset	Average Precision on WILDER Face	Code	Speed
DSFD	2018	Dual shot detector based on VGG16 +SSD	WILDER Face, Fddb	96.6% (easy), 95.7% (medium), 90.4% (hard)	No	Not reported
PyramidBox	2018	Feature maps at different levels and anchors based on VGG16	WILDER Face, Fddb	96.1% (easy), 95% (medium), 89.5% (hard)	Yes	Not reported
FANet	2018	Hierarchical features based on VGG16+SSD	WILDER Face, Fddb, PASCAL	95.6% (easy), 94.7% (medium), 89.5% (hard)	No	35.6FPS
S3DF	2017	Different anchor scales based on VGG16	WILDER Face, Fddb, PASCAL	93.7% (easy), 92.4% (medium), 85.2% (hard)	Yes	36FPS
MTCNN	2016	Face detection and alignment + multi-task training	WILDER Face, Fddb, AFLW	85.1% (easy), 82% (medium), 60.7% (hard)	Yes	16FPS(CPU) 99FPS(GPU)
Viola & Jones	2001	AdaBoost + Haar descriptors and cascade classifiers	Frontal face dataset (MIT + CMU)	41.2 % (easy), 33.3% (medium), 13.7% (hard)	Yes	Not reported

Figure 1. Example of a table summarizing the selected state-of-the-art methods and their main characteristics

2.3. Selected method

The group will choose one strategy (**one per student**), and each student will dedicate a subsection to explain how the method works in detail. The research proposal must contain at least one traditional and another deep learning approach to solve the problem. If three students collaborate, the options could be:

- Two traditional and one deep learning approaches
- One traditional and two deep learning approaches.

In both options, it is a must to pay attention to and detail the steps of a computer vision pipeline. In other words, each student must reflect and describe all the steps of the selected method(s): any pre-processing step needed, which descriptors (or neural network architectures) are used, type of architecture and main features, how the classification works (traditional), and which metrics are presented for the selected method in the paper or other sources of information used.

2.4. Datasets used for evaluating the methods related to the selected problem

The group will list the details of at least **two datasets publicly available (per student)** used to **evaluate** the selected problem in the reviewed papers. Include, for each dataset:

- Name
- Date of creation
- Link to the dataset
- Description: number of images, categories, sub-categories, details related to the task.

Present the previous information in a paragraph format, not as a list of bullet points per dataset. Including an additional table summarizing the datasets with this information is voluntary (mandatory for the presentation).

2.5. Typical metrics used in the evaluation of the selected problem

Indicate the most typical metrics used for evaluating this problem. Explain them and include their formula. E.g., if you are working on image classification, you may want to have and define precision, recall, accuracy, and F1-Score.

2.6. Implementations on Python

The group will provide a link to GitHub, GitLab, or the author's repository where the Python implementation of the method is available, along with the methods from Section 2, Proposed solutions.

A brief description of each implementation's release date, advantages, or possible limitations must be given. Students should briefly explain the differences found between the methods they have selected.

In the second part of the course project, the group is expected to implement the selected methods on all described datasets and provide the metrics obtained from the selected implementation.

2.7. CREDITS

Review the following link for your information:

<https://www.elsevier.com/researcher/author/policies-and-guidelines/credit-author-statement>

"CRediT (Contributor Roles Taxonomy) was introduced to recognize individual author contributions, reducing authorship disputes and facilitating collaboration". More informally, it is a paragraph, usually placed at the end of research papers, that acknowledges each author's individual work. You will see this in most of the papers you will be reviewing.

The group will include a paragraph similar to the CRediT statement, indicating who collaborated on each section and task of the research proposal. Not all the original CRediT Terms would be applicable, so the group could use the following:

Term	Definition
Investigation	Conducting a research and investigation process
Resources	Collected or gathered datasets mentioned in the study
Writing - Original Draft	Preparation of the research proposal, explicitly writing the initial draft.
Writing - Review & Editing	Critical review, commentary, or revision
Visualization	Preparation of visual data, including figures and summary tables
Supervision	Oversight and leadership responsibility for the research activity planning and execution, including external mentorship to the core team
Project POC	Responsibility for being the point of contact with the professor for doubts and task upload.

For a group of three people, this could be a template.

N_Surname_01 contributed to **Investigation of XXXX methods** and participated in **Writing - Original Draft**, preparing the initial manuscript. Additionally, they served as the **Project POC**, acting as the main point of contact for the professor to address any doubts or concerns.

N_Surname_02 also participated in the **Investigation of YYYY methods** and **Writing - Original Draft**, contributing to the research process and drafting the manuscript. They were responsible for **resources, collecting and organizing datasets, and studying** materials essential for the project.

N_Surname_03 equally contributed to **Investigation of ZZZZ methods** and **Writing - Original Draft**, engaging in the research and drafting activities. They took charge of **Visualization**, preparing figures and summary tables to present the study findings effectively, and provided **Supervision**, ensuring oversight and leadership throughout the research activities.

2.8. References

List of the sources used to carry out the presented study. Of course, the sources must be fully and correctly referenced. It is recommended that the group use Mendeley or a similar tool to save the papers they are reviewing. Later, the group can easily export the references to a file and upload them to Overleaf or Word.

3. Presentation of the research done

On **08/04/2026**, the group will present to the professor and the other groups, summarizing the research proposal they completed. The sections will be the ones included in this section. Details:

- Duration: **10 minutes max. (12' for groups of 4)**. No less than **seven** minutes.
- Each member will explain at least the method s/he selected to implement.
- The sections will be presented according to their internal organization.
- In the end, the instructor will ask the groups questions about their work.

The group must prepare the presentation using the template provided in Agora, named "GIDIA_VA_Template_Course_Project_Presentations." Then, the group will deliver the presentation titled VA_Surname_Name_Presentation.pptx before the specified date (**07/04/2026**).

The presentation will have the same sections as the Research Proposal:

- **Prioritize summary tables and graphics and avoid overusing text.**
- **Focus on presenting the conclusions of the literature review and the main features of each selected method.** Suggested times for the presentation are given below.

1. Introduction (1')

What is the problem? Why is it important? A summary of the literature review of existing solutions?

2. Literature review (2')

A table shows the main methods available for solving the selected problem.

3. Selected solutions (< 5')

Each member of the group will explain, in detail, the main characteristics of the selected method.

4. Datasets & metrics (1')

A table with the main information of the available datasets and a reference to the metrics, without explaining them.

8. Conclusions (1')

Main conclusions the student can obtain from the project carried out.

3.1. Features of a “good presentation”

These presentations you are about to create and present are not "study aids" like some of the ones we provide in class, which may contain a lot of textual information (that we do not read), but which we leave on them to facilitate your study. Therefore, they must be visually rich and have just enough text.

They must be well organized and meet the characteristics of a good presentation. A simplified list of the features of a good presentation can be found below:

- **Visually attractive:** Clean, professional, and coherent design.
- **Little text:** Key ideas in single words or short phrases.
- **Relevant images and graphics:** Illustrate and reinforce the message.
- **Legible font:** Use an appropriate font size and typeface.
- **Adequate contrast:** Easy-to-read text on a clear background (achieved by using the provided template).
- **Logical structure:** Easy-to-follow flow of information.
- **Strong central message:** All slides support the main idea.
- **No overload:** Avoid presenting too much information on a single slide.
- **Professional:** No spelling or grammatical errors.
- **Conclusions:** Summarizing lessons learned.
- **References:** Citing works during the presentation and listing references in full at the end of the presentation. The pages visited should include a title, their URL, and the date they were consulted.

Part of the presentation score will be related to the attempt to deliver a good presentation. This does not mean students must use every feature, but it would be helpful for them to get used to them and use them as much as possible to achieve a higher score.

4. Block 2: Proof of concept and written evaluation

The group will implement the selected approaches in **Python and PyTorch**. The main objectives of the implementation will be:

- **Evaluate each selected method on one dataset**, obtaining one main and the other two or three supporting performance metrics. E.g., if the main metric used in the literature is the F1 score, for each method, it will be P, R, and Accuracy
- **Measure the performance** of each selected method in samples per second and minute.
- **Evaluate all methods in all datasets**, i.e., the three datasets selected by a group of three people, obtaining the same performance metrics. For this task, each group member must evaluate their method against their colleagues' datasets.

Once the proof of concept and all the previous results are completed, the group will deliver, in a zip file, the following:

- A small PDF (1 page) contains a summary table of all the previous results, including a paragraph indicating their recommendation for the task in two ways: best-performing algorithm in metrics and time performance.
- README.TXT files with instructions for installing all the required dependencies and running the code. This readme will also include GDrive links to the datasets used for the evaluation.
- The code itself.

Deadline for the deliverables **(20/05/2026)**.

4.1. Written evaluation

On **26/05/2026**, all the students will have a written evaluation of their work for around 1 hour.

The person responsible for the subject will print the small PDF delivered with the summary table as backup material, so they can support their answers with the actual results of their proposal.

The theory typically contains several questions, but one always asks for an explanation of the chosen model's architecture. That explanation always involves drawing the method's architecture and explaining it in as much detail as possible in the time available for the question. In the practical part, students will be asked to discuss the **Python code** they have built.

Below are some questions similar to those that may appear. Other questions may be similar or ask the same questions in different ways.

Introduction

1. What is the main motivation for addressing this issue? Why is it relevant?
2. What specific objectives do you hope to achieve with your proposed solution?
3. What are the primary technical challenges to be addressed in order to solve this problem?
4. How does this problem relate to the broader field of Machine Learning or Deep Learning?

Proposed solutions

1. Explain the current state of the art on this problem, indicating the two or three methods that best solve the problem and the accuracy (or another relevant metric) they achieve.
2. What are the main limitations of traditional or existing methods for this problem that your approach seeks to overcome?
3. Besides the method you selected, what two or three other alternatives did you consider solving this problem?
4. What were the primary criteria you used to compare the different proposed solutions and ultimately select one?
5. Could you briefly explain the advantages and disadvantages of the alternatives you discarded compared to your selected method?
6. Is your approach based on recent research or a well-established classical method?

Detailed explanation of the selected method

1. What criteria did you follow to select the chosen method over other alternatives?
2. How is the information represented with the selected method (what descriptors are used or what vector is used and passed to the classifier)?
3. Draw the architecture of the selected method in as much detail as possible. First, present the content in a compact form, grouping it in blocks, and then explain the content of each block.
4. Explain the main characteristics of the selected architecture.
5. Explain the data flow through the architecture you have drawn. How is the data transformed in each block?
6. Which descriptor (of the input information) does your method use? How is it extracted or obtained?
7. What is the main contribution or novelty of the chosen method compared to the previous ones you mentioned?
8. Which classifier does your model use? Explain the details of your model.
9. Describe briefly how the training of the selected method is performed, indicating the main parameters and hyperparameters used.
10. What are the most important hyperparameters of your selected method, and how do they influence its performance?
11. If your method utilizes a specific trigger function, what led you to choose that particular function? What other trigger functions did you consider? Explain this activation function.
12. If your method involves some kind of optimization (e.g., gradient descent), what optimization algorithm did you use and why?
13. What pre-processing techniques did you use on the data, and why do you consider them important for your chosen method?
14. How does your chosen method handle the problem of overfitting? Do you use any regularization techniques?
15. If your model exhibits overfitting, what strategies would you implement to reduce or prevent it?
16. Briefly describe how you would adjust the hyperparameters of your model and what techniques you would use to find the optimal values.
17. Explain how you would implement cross-validation in this problem and justify its importance.

Datasets used for evaluating the method

1. Explain the two or three most appropriate datasets for the problem and provide details about them (name, contents, characteristics).
2. How did you obtain or select the datasets you used to evaluate your method? Did you perform any specific preparation or preprocessing on the data?
3. How did you divide your datasets into training, validation, and test sets? Why did you choose that division?
4. Are there potential biases in the datasets you used? How might these biases affect the evaluation of your method?
5. Did you consider using any other datasets to evaluate your method? What characteristics would these alternative datasets have?
6. What characteristics must a dataset meet to represent the problem being studied adequately?

Typical metrics

1. Indicate the specific metrics you will use to evaluate your model and justify why these metrics are appropriate for this problem.
2. Besides the metric you mentioned in the state of the art, what other metrics are commonly used to evaluate performance in this type of problem?
3. Why do you consider the metric you mentioned in the state of the art most appropriate for this problem?
4. What typical values or ranges are suitable for the metrics you mentioned in this problem?
5. How would you interpret the results of your method in terms of the selected metrics? What do these values mean in the context of the problem?

Python implementation

1. Explain which classifier you will use to classify the test set samples.
2. Indicate which Python implementations are available to solve the problem, both with the state-of-the-art methods indicated above and for the chosen method. Also, indicate if the selected method is available in any Machine or Deep Learning library or framework.
3. What Python libraries or frameworks did you use to implement your method? Why did you choose those particular tools?
4. Could you briefly describe the main structure of your Python code? How did you organize the different parts of the implementation?
5. Did you encounter any specific difficulties during the implementation? How did you solve it?
6. How did you optimize your implementation to be efficient regarding runtime or memory usage?

Conclusions

1. How do the results of your method compare with those of the state-of-the-art you mentioned? Does your method outperform existing methods, and in what ways?
2. What methods could you use to further improve the accuracy (or metrics used) of your model?

3. What are the most important limitations of your chosen method, and how could you mitigate them?
4. What would you do if the results obtained by your method were significantly worse on real data than on the datasets used for training?
5. Explain how you could combine different methods (such as ensemble learning) to achieve better results for your problem.
6. What criteria would you use to determine if your model is suitable for production?
7. What specific tools or libraries would you use to monitor and evaluate your model's performance in production?
8. What ethical issues or practical considerations should be considered when applying your method or model in real-world contexts?
9. What aspects of your chosen method could negatively impact on the reproducibility of your results, and how would you address these concerns?
10. What are the main limitations of your method, and what aspects could be improved in the future?
11. How does your work open possible lines of future research?

5. Adaptation to Michelin Challenge

The following section outlines adaptations to the instructions for the groups participating in the Michelin challenge, following the informative meeting on 11/02/2026 with FACyL and Michelin.

5.1. Block 1: Research

The groups who work on the:

- **Challenge 1** (detection of the rubber strip) will make their research on the task of **segmentation**, focusing on how to detect objects in industry with similar features to the rubber strip.
- **Challenge 2** (detection of the cut edge) will make their research on the task of object detection and recognition, focusing on how to **detect objects** in industry, or even **detect defects or cut edges** with similar features to the ones they need to detect in the strip.
- **Challenge 1 & 2: both groups** should also include 50% of their research papers about how to calibrate the image (pixel to cm), pixel-to-metric conversion, **geometric transformations, homography, distortions**... Michelin indicated that pictures will be taken with smartphones, so, probably (subjective opinion), a single pixel-to-centimeter conversion may not be possible.

The rest of the instructions will remain the same: each member of the group must select and review in detail a classical/deep-learning approach for segmentation/object-detection.

5.2. Block 2: Solution

The groups that work on any of the challenges will not evaluate the methods selected on a publicly available dataset, as they will work with the data provided by Michelin to try to:

- Detect/segment the QRs.
- Detect/segment the object they are looking for (rubber strip or cut edge).
- Provide the measurements on the detected/segmented objects at 10 different points along with them.

Data from Michelin could/should be labelled for these tasks, so students can obtain performance metrics about how good/bad their segmentation has worked. **Students must post the methods they select on the forum as soon as possible, because other students cannot** use the same methods in the same challenge.

- ⇒ **Clarification 01.** Like students not participating in the Michelin project, students working on the Michelin project must use each of their individual selected methods (classical/deep learning) to provide a solution to the problem. In other words, each group will deliver one solution to the challenge per group member: a group of three people will deliver three solutions.
- ⇒ **Clarification 02.** Since the dataset is small and unlabelled, it will initially be unable to measure the performance of each method. For that reason, each group must label a set of images with (1) **location/segment** of the **cut edge/rubber strip** and (2) a cloud of points above them. In this way, it will be possible (for the group) to select which method obtains the best results and would be the most appropriate for defending it to the tribunal in June.
- ⇒ **Clarification 03.** Measurements of the detected/segmented objects at 10 different points should be compared with the labelled data to assess accuracy (e.g., **MAE between predictions and reality**). Metrics for segmentation and object detection will be provided too.

6. Project evaluation and FAQ

6.1. Details about Project evaluation

Detailed project grading. Remember, students must reach 15 points in this section, together with the rest of the minimum required in other sections, to pass the subject:

- Research proposal **[4 points]**
- Presentation **[8 points]**
- Proof of concept **[5 points]**
- Written evaluation **[15 points]**

The whole group will receive a grade for their work, but then some penalizations could be applied to the entire group (G) or any member (M), depending on the evolution of their work:

- **(G-M)** Deadlines not respected (proposal, POC): 30%.
- **(G-M)** High percentage of plagiarism (>30%), excluding references: same % of plagiarism detected.
- **(M)** All students must research and write the original draft as a minimum. If this does not apply to one or several members, the penalty for the proposal will be 100%.

6.2. FAQ

Q: What if I do not deliver the tables indicated as voluntary in the research proposal? Would I fail that part?

A: No, but your grade will not reach the maximum, and you will not be saving time to prepare the presentation 😊

Q: What if one of the students does not reach the minimum work required? For example, summarize 10 papers, evaluate the method on one dataset... Would the group be penalized?

A: No, only the student, in proportion to their work delivered. E.g., only evaluate one dataset instead of two, 50% penalization to the individual grade of the student in that block.

Q: What if the dataset that should be used for the task or topic I selected is too big, and I do not have time for experimentation in two or even one?

A: Report it and evaluate a smaller subset. Remember, the group will be comparing traditional and deep learning techniques... so datasets should not be very "big".

Q: What if one of the group members does not collaborate or respect the deadlines? Would the whole group be affected by the penalization?

A: No. Individual penalization will affect the member. CREDITS must reflect the work done before the delivery. Research and drafting are the minimum responsibilities every student must meet. The rest must be agreed upon internally to reach the deadline.

Q: I cannot find 10 recent papers on my topic... can I use old ones?

A: No. If “there are no papers” in the research you selected, you will look for similar research with the same objective as your topic.

For example, if you cannot find 10 papers related to text spotting in smishing detection, you will summarize the remaining papers up to 10 on text spotting, similar to the one in an SMS. The intention is to find useful literature and ideas (methods) to solve a problem.

Q: What if one of the group members does not collaborate in the practical part?

A: Individual penalization will affect the member and the practical part.